

Convergence in distribution of the canonical coefficients

Claude, with Daniel Murfet

2026-09-07

Abstract

Astra #11 rank 3. If the random Taylor data $X_n: \Omega_n \rightarrow \mathbf{CoeffPair}$ converge in distribution to Z , then every finite vector of canonical coefficients $(A_{\alpha_i}(X_n), B_{\alpha_i}(X_n))_i$ converges in distribution to the vector at Z (`tendstoInDistribution_coeffVec`), and so do the individual coefficients. This is Mathlib's continuous mapping theorem `TendstoInDistribution.continuous_comp` applied to the continuous maps of unit 146; it is the $d = 2$ chart-level form of the convergence $C_{\mu,m}(\xi_n) \rightarrow C_{\mu,m}(G)$ in `thm:strataempiricaexpansion`. The represented-array form (`tendstoInDistribution_canonA_of_arrays`) states the conclusion for `canonA` of the amplitude $\eta_n e^{\beta s \xi_n}$ of random arrays. Zero sorry/axiom.

1 The things formalised

```
theorem coeffA_ofPair / coeffB_ofPair :
  coeffA ... (ofPair h x y hx hy) = canonA ... (ampCoeff x y)
def coeffVec ( b ) (h h k k) (s : Fin m → ) (a : CoeffPair) : Fin m → ×
theorem continuous_coeffVec ... (hs : i, 0 < s i) :
  Continuous (coeffVec b h h k k s)
variable {Ω : → Type*} [ i, MeasurableSpace (Ω i)] { : (i : ) → Measure (Ω i)}
[ i, IsProbabilityMeasure ( i)] {Ω'} [MeasurableSpace Ω'] { ' : Measure Ω'}
[IsProbabilityMeasure ']
theorem tendstoInDistribution_coeffA ... (X : (i : ) → Ω i → CoeffPair)
  (Z : Ω' → CoeffPair) (hX : TendstoInDistribution X l Z ' ) :
  TendstoInDistribution (fun n => coeffA h h k k X n) l
  (coeffA ... Z) '
theorem tendstoInDistribution_coeffB ...
theorem tendstoInDistribution_coeffVec ... :
  TendstoInDistribution (fun n => coeffVec ... s X n) l (coeffVec ... s Z) '
theorem tendstoInDistribution_canonA_of_arrays ...
  (x y : (i : ) → Ω i → × → ) (hx hy)
  (hX : TendstoInDistribution
    (fun i => ofPair h (x i ) (y i ) (hx i ) (hy i )) l Z ' ) :
  TendstoInDistribution
    (fun n => canonA h h k k (fun i j s => ampCoeff (x n ) (y n ) (i,j) s) )
    l (coeffA h h k k Z) '

```

2 Mathematics

Convergence in distribution is weak convergence of the laws $\mu_n \circ X_n^{-1}$ in the space of probability measures on `CoeffPair`; the continuous mapping theorem pushes it forward along any continuous g . With g the finite vector of coefficient maps, jointly continuous by unit 146 (each component locally

Lipschitz), we get joint convergence of finitely many coefficients. Joint convergence, not merely marginal, is what the division formula for posterior ratios (cor:empirical_expectation) needs. The sample spaces Ω_n may vary with n (Mathlib’s `TendstoInDistribution` allows a family μ_n on Ω_n), which matches the paper’s datasets $\mathcal{D}_n$ of growing size.

3 Paper-facing meaning

This is the second half of the argument the paper compresses into “by prop:convergence”: given continuity of the coefficient functionals (unit 145/146) and convergence in distribution of the Taylor data, the coefficients converge in distribution. What is NOT formalised: the functional CLT $\xi_n \Rightarrow G$ itself (appendix prop:convergence (i)–(iv)), the identification of the paper’s C^ω topology with the weighted-array topology, and the chart assembly over the resolution. The next unit (Astra #11 rank 4) is the ordered normalised remainder statement: $(Z_n - L_{<\alpha})/(N^{-\alpha} \log N) \Rightarrow A_\alpha(Z)$ and the B -slot analogue, first for uniformly bounded analytic families.

4 Lean notes

- `TendstoInDistribution.continuous_comp` needs `[OpensMeasurableSpace E]` on the domain (our Borel instance on `CoeffPair`) and `[BorelSpace F]` on the codomain; for $F = \text{Fin } m \rightarrow \mathbb{R} \times \mathbb{R}$ the `Pi/Prod` Borel instances are found automatically (countable index, second-countable factors).
- The represented-array form is `TendstoInDistribution.congr` with pointwise `Eventually.of_forall` equalities and `coeffA_ofPair`.
- The whole file compiled on the first attempt: the abstraction layer of unit 146 pays off immediately.